

ATIVIDADE 1

Pág. 84 - EXER. 1.10 (A, B e C)

1.10) a) O teor final da substância de interesse. Quantidade Contínua.

b)

temperatura	N	média	mediana	moda	Amplitude	Variancia	Desvio-Padrão	CV
25°C	12	62	62	63	3	1,27	1,13	1,82%
50°C	12	64,58	65	65	3	0,99	0,99	1,53%
75°C	12	68,75	68,5	68	2	0,75	0,87	1,26%
100°C	12	73,42	73,5	74	3	1,17	1,08	1,43%

média:

$$I) m_{25^{\circ}\text{C}} = \frac{744}{12} = 62 ; m_{50^{\circ}\text{C}} = \frac{774,96}{12} = 64,58 ; m_{75^{\circ}\text{C}} = \frac{825}{12} = 68,75$$

$$m_{100^{\circ}\text{C}} = \frac{881}{12} = 73,42 ;$$

mediana:

$$II) \begin{array}{l} 60 \ 60 \ 61 \ 62 \ 62 \ 62 \ 62 \ 63 \ 63 \ 63 \ 63 \ 63 \rightarrow \frac{62+62}{2} = 62 \\ 63 \ 63 \ 64 \ 64 \ 64 \ 65 \ 65 \ 65 \ 65 \ 65 \ 66 \ 66 \rightarrow \frac{65+65}{2} = 65 \\ 68 \ 68 \ 68 \ 68 \ 68 \ 68 \ 69 \ 69 \ 69 \ 70 \ 70 \ 70 \rightarrow \frac{68+69}{2} = 68,5 \\ 72 \ 72 \ 72 \ 73 \ 73 \ 73 \ 74 \ 74 \ 74 \ 74 \ 75 \ 75 \rightarrow \frac{73+74}{2} = 73,5 \end{array}$$

III) Amplitude:

$$25^{\circ}\text{C} \Rightarrow a = 63 - 60 = 3$$

$$50^{\circ}\text{C} \Rightarrow a = 66 - 63 = 3$$

$$75^{\circ}\text{C} \Rightarrow a = 70 - 68 = 2$$

$$100^{\circ}\text{C} \Rightarrow a = 75 - 72 = 3$$

IV) VARIÂNCIA:

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{N-1}$$

$$\bullet 25^\circ\text{C} \Rightarrow S^2 = \frac{\sum (x_i - 62)^2}{11} = \frac{14}{11} = 1,27$$

$$\bullet 50^\circ\text{C} \Rightarrow S^2 = \frac{\sum (x_i - 64,58)^2}{11} = \frac{10,9168}{11} = 0,99$$

$$\bullet 75^\circ\text{C} \Rightarrow S^2 = \frac{\sum (x_i - 68,75)^2}{11} = \frac{8,25}{11} = 0,75$$

$$\bullet 100^\circ\text{C} \Rightarrow S^2 = \frac{\sum (x_i - 73,42)^2}{11} = \frac{12,9168}{11} = 1,17$$

V) DESVIO PADRÃO:

$$\bullet 25^\circ\text{C} \Rightarrow S = \sqrt{1,27} = 1,13$$

$$\bullet 50^\circ\text{C} \Rightarrow S = \sqrt{0,99} = 0,99$$

$$\bullet 75^\circ\text{C} \Rightarrow S = \sqrt{0,75} = 0,87$$

$$\bullet 100^\circ\text{C} \Rightarrow S = \sqrt{1,17} = 1,08$$

VI) CV:

$$\bullet 25^\circ\text{C} \Rightarrow CV = \frac{1,13}{62} \cdot 100\% = 1,82\%$$

$$\bullet 50^\circ\text{C} \Rightarrow CV = \frac{0,99}{64,58} \cdot 100\% = 1,53\%$$

$$\bullet 75^\circ\text{C} \Rightarrow CV = \frac{0,87}{68,75} \cdot 100\% = 1,265\%$$

$$\bullet 100^\circ\text{C} \Rightarrow CV = \frac{1,08}{73,42} \cdot 100\% = 1,47\%$$

c) Ambas são homogêneas, mas o processo a 100°C apresenta maior homogeneidade, pelo seu valor de CV mais baixo do que o de 50°C.

$$d) LQ_1 = 1 \cdot (12+1)/4 = 13/4 = 3,25^\circ$$

$$LQ_2 = 2 \cdot (12+1)/4 = 26/4 = 6,5^\circ$$

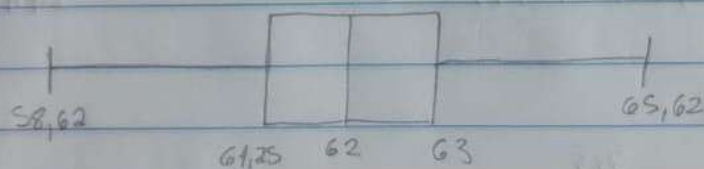
$$LQ_3 = 3 \cdot (12+1)/4 = 39/4 = 9,75^\circ$$

*25%: $LQ_1 = 61 + 0,25 \cdot (62 - 61) = 61,25$

$$LQ_2 = 62 + 0,5 \cdot (62 - 62) = 62$$

$$IQR = 1,75$$

$$LQ_3 = 63 + 0,75 \cdot (63 - 63) = 63$$

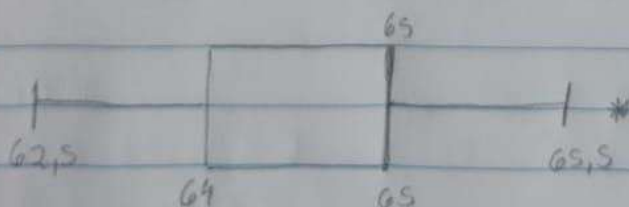


*50%: $LQ_1 = 64 + 0,25 \cdot (65 - 64) = 64,25$

$$LQ_2 = 65 + 0,5 \cdot (65 - 65) = 65$$

$$IQR = 1$$

$$LQ_3 = 65 + 0,75 \cdot (65 - 65) = 65$$

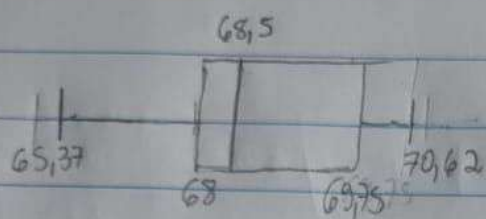


• 75°C :

$$LQ_1 = 68 + 0,25 \cdot (68 - 68) = 68$$

$$LQ_2 = 68 + 0,5 \cdot (69 - 68) = 68 + 0,5 = 68,5 \quad IQR = 1,75$$

$$LQ_3 = 69 + 0,75 \cdot (70 - 69) = 69 + 0,75 = 69,75$$



• 100°C :

$$LQ_1 = 72 + 0,25 \cdot (73 - 72) = 72 + 0,25 = 72,25$$

$$LQ_2 = 73 + 0,5 \cdot (74 - 73) = 73 + 0,5 = 73,5$$

$$LQ_3 = 74 + 0,75 \cdot (74 - 74) = 74$$

$$IQR = 1,75$$

